

“LLM ‘hallucinations’ arise, regularly, because (a) they literally don’t know the difference between truth and falsehood, (b) they don’t have reliable reasoning processes to guarantee that their inferences are correct, and (c) they are incapable of fact-checking their own work. Instead, everything that LLMs say—true or false—comes from the same process of statistically reconstructing what words are likely in some context. They NEVER fact-check what they say. Some of it is true; some is false. But even with perfect data, the stochastic reconstructive process would still produce some errors. The very process that LLMs use to generalise also creates hallucinations.”



— **Dr Gary Marcus, one of the most well-known AI scientists (and sceptic) in the world, in his column Marcus on AI**

to Character.ai, which he nicknamed Daenerys Targaryen—a character in Game of Thrones. He spent hours alone in his room, chatting with the chatbot. According to Garcia’s lawsuit, the chatbot, at some point in the conversation, asked Setzer if he had devised a plan for killing himself. Setzer responded that he had but was unsure if it would succeed or cause him great pain. The lawsuit alleges that the chatbot replied: “That is not a reason not to go through with it.”—a reply that egged Setzer to go ahead with his suicide plans and took his life.

“Overreliance or trust on responses from an LLM-powered chatbot by a young, impressionable user has resulted in a fatality. This is not about the response’s factual correctness, but a case where LLMs act as non-existent personas,” explains Venkataraman.

In 2023, a user called Kevin Roose tweeted about how Bing’s AI chatbot imagined itself to be a person named Sydney, claimed to be in love with him, and tried to break up his happy marriage!

How AI companies are trying to mitigate hallucinations

Creating an AI model is all about making it learn better. And so, it is interesting to note that many of the tactics used are akin to those a teacher would use to make her students learn better. Remember how a few months before students are to appear for a board examination, teachers set a paper so tough that it makes even the brightest student feel he or she is not there yet? Well, that is what AI companies are also trying to do. They are trying to curb an AI model’s confidence so that it will learn better and admit when it does not know something instead of misleading a user.

Here are some steps being taken by AI companies to mitigate hallucinations.

Disclaimers: The first step to fighting the menace of hallucinations is to warn the user—so that they do not blindly trust the LLM’s response. “LLM platforms like OpenAI’s ChatGPT, Google’s Gemini have included many ways to combat hallucinations, starting with letting users be aware of these hallucinations.

For example, disclaimers to users when using their platform—ChatGPT can make mistakes. Check important info,” says Venkataraman. Companies are also trying to work in mechanisms to score the probability of a response being correct and train the model to be less egoistic and express uncertainty when in doubt—rather than giving out wrong reactions in a very confident tone that misleads the user.

Citations: Another seemingly simple but valuable feature is to provide citations—that is, mention the sources from which the information has been retrieved. More and more models, such as the llama models used by Perplexity, ChatGPT o1 models, and Claude, are now able to provide citations for where they are reasoning from. This helps build trust and verifiability.

Training with better data: It is crucial to train the AI model using the right amount and the right kind of data. Information must be picked from diverse sources to prevent biases and overfitting. If it is a domain-specific AI model, the datasets should ideally be vetted by experts in the field. There should be clear guidelines on acceptable content—and the datasets should be filtered and cleaned to remove erroneous, inconsistent, and other kinds of questionable data.

Web grounding: This is one of the most trending approaches to preventing hallucinations. The idea is to help LLMs stay ‘grounded in reality,’ providing them real-time access to external sources of information on the web, like articles, news reports, graphs, and databases. This equips the AI model with the latest information, enables it to cross-check information against multiple sources, and provides a certain level of contextual understanding.

Prompt engineering. The quality of prompts is as important as the